

Sharp denominator corrections for Brown–Zudilin cellular linear forms in $\zeta(5)$: measurements, a conjecture, and a lattice mechanism

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Abstract

Brown and Zudilin [arXiv:2210.03391] constructed effective rational approximations $I'(\mathbf{a} \cdot n) = Q_n \zeta(5) - P_n$ to $\zeta(5)$ from cellular integrals on $\overline{\mathcal{M}}_{0,8}$, together with experimental integrality statements for the coefficients. By recovering the exact rational coefficients $P_n, \hat{P}_n$ at more than 200 parameter points via certified integer-relation computations, we show that the published integrality claims are false as stated—their own printed values violate them—and that the correct statements carry a correction factor of exactly $12 = 2^2 \cdot 3$ (respectively 2 for the $\zeta(3)$ -companion), which we conjecture to be sharp and verify in every measured cell. Unlike the familiar correction factors in the classical literature (Apéry, Rhin–Viola, Krattenthaler–Rivoal), which we check are uniformly *removable*, this correction is *attained*. We propose a mechanism: the coefficients are periods against a non-primitive integral Betti class produced by the construction’s motivic elimination of $\zeta(2)$, so the correction is a lattice index bounded by the Bernoulli constant 24 arising from $\zeta(2) = -(2\pi i)^2/24$; an elementary elimination-cost lemma, a conditional theorem, and a concrete derivation in the totally symmetric case (where the factor 12 appears as a doubled sine-kernel Laurent coefficient $\frac{1}{6}$) support this. As by-products we determine the supremum of the Brown–Zudilin “worthiness” exponent over the full parameter cone, verify that the leading factor 2 in a denominator conjecture of Krattenthaler–Rivoal is unnecessary for $n \leq 5$, and document an orbit-coherent boundary-prime anomaly. All measured denominator slack is bounded; no measured direction offers exponential arithmetic gain.

1 Introduction

For the linear forms of Apéry-type used in irrationality proofs, the arithmetic of the rational coefficients is traditionally expressed by *inclusions*: statements of the shape $D_n \cdot (\text{coefficient}) \in \mathbb{Z}$ for explicit denominators D_n built from $d_n = \text{lcm}(1, \dots, n)$. These inclusions are proved (or conjectured) from term-by-term analysis, and the question of their *sharpness*—whether the true denominators actually attain D_n —is rarely addressed and, to our knowledge, has never been investigated systematically by direct computation.

This paper does so for the family of cellular integrals $I(\mathbf{a})$ on $\overline{\mathcal{M}}_{0,8}$ introduced by Brown and Zudilin [1], which produce simultaneous rational approximations $I' = Q\zeta(5) - P$, $I'' = Q\zeta(3) - \hat{P}$ by a motivic elimination of the parasitic $\zeta(2)$. Our findings:

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1. The integrality statements printed in [1] (their experimental inclusions for I' , and the totally symmetric display $d_n^5 P_n \in \mathbb{Z}$) are **false as stated**, already against the paper’s own printed values (§3).
2. The corrected statements carry a factor of exactly 12 (resp. 2), and this factor is **sharp**: it is attained in 46 of 164 systematically sampled cells at the prime 2 and 30 cells at the prime 3, and never exceeded in any of our more than 200 certified cells (§4). One isolated, orbit-coherent, non-persistent exception at a boundary prime $p = 7$ is documented in §5.
3. By contrast, the analogous correction factors appearing in the classical literature are uniformly *removable*: we verify $d_n^3 a_n \in \mathbb{Z}$ for Apéry’s $\zeta(3)$ forms for $n \leq 50$ (sharpening the ubiquitous “ $2d_n^3$ ”), and we verify that the leading factor 2 retained even in the *conjectured* sharp denominator law of Krattenthaler–Rivoal [3, eq. (17.1)] for Zudilin’s $\zeta(5), \dots, \zeta(11)$ forms is unnecessary for $n \leq 5$ (§8)—apparently the first direct computation of those coefficients.
4. We propose a mechanism (§6): the elimination of $\zeta(2)$ forces the linear form to be the period of a *non-primitive* integral Betti class, and the identity $\zeta(2) = -(2\pi i)^2/24$ prices any integral elimination at a lattice index dividing 24. An elementary lemma makes this precise; the measured constants 12 and 2 then *measure* 2-adic indices of the integral Betti lattice of the underlying rank-3 mixed Tate motive. In the totally symmetric case we exhibit the factor concretely: the excess enters through the doubled Laurent coefficient $\frac{1}{6}$ of the reciprocal-sine kernel, and we prove unconditionally that no prime $p \geq 5$ contributes (§7).
5. All measured denominator slack (cells where the true denominator is *smaller* than predicted) is bounded along every measured direction, including growth series $n \leq 8$; consequently the construction’s “worthiness” budget receives no hidden arithmetic subsidy. We also determine $\sup \gamma = 0.86597135\dots$ over the full 7-dimensional projective parameter cone—attained exactly at the authors’ published point, whose symmetric coordinates form an arithmetic progression (§9).

Throughout, “verified” means: exact integers Q_n from the closed binomial double sum of [1]; values of $I(\mathbf{a} \cdot n)$ computed to a certified precision (hundreds of digits) from the 1-dimensional integral of a product of two ${}_3F_2$ ’s given in [1], evaluated by an exact partial-fraction/polylogarithm method; and $P_n, \hat{P}_n$ recovered by PSLQ over the basis $\{1, \zeta(2)\}$ with residual certification, precision-stability checks, and exact agreement with all rational values printed in [1] (§2).

2 The family and the measurement pipeline

We follow the notation of [1]. For admissible integer parameters $\mathbf{a} = (a_1, \dots, a_8)$ (the 17 convergence forms nonnegative; we require all 28 forms of the multiset $\mathbf{h}(\mathbf{a})$ nonnegative so that the full symmetry orbit is admissible), the cellular integral decomposes as

$$I(\mathbf{a}) = 2Q(\zeta(5) + 2\zeta(3)\zeta(2)) - 4\hat{P}\zeta(2) - 2P, \quad Q \in \mathbb{Z}, P, \hat{P} \in \mathbb{Q},$$

and the object of interest is $I' = Q\zeta(5) - P$. The group $\mathfrak{S} \cong \Sigma_7$ acts on normalized integrals through the symmetric parameters $s_0; s_1, \dots, s_7$, permuting the 28-element multiset $\mathbf{h}$, with $I(\mathbf{a})/\prod_{i \in \mathcal{F}} h_i!$ invariant for the 17-element index set $\mathcal{F}$ of [1].

Exact Q. $Q(\mathbf{a} \cdot n)$ is computed exactly from the double binomial sum [1, eq. (sumQ)].

Certified I. $J(\mathbf{p}; \mathbf{q}) = I(\mathbf{a})$ reduces [1] to a 1-dimensional integral of a product of two ${}_3F_2$'s. For integer parameters each ${}_3F_2$ coefficient is a proper rational function of the summation index; its exact partial fractions over $\mathbb{Q}$ convert every evaluation into finitely many polylogarithms $\text{Li}_m(z)$, $z \leq 0$, with rational weights (analytic continuation built in). Working precision is raised until two runs agree; tanh–sinh quadrature errors are certified against the target precision. On validation, this evaluator agrees with the direct (slow) hypergeometric path to all compared digits (≥ 40) and is $\sim 57\times$ faster on production cells.

Recovery. With ζ -values to matching precision, $x = (2Q\zeta(5) + 4Q\zeta(3)\zeta(2) - I)/2 = P + 2\hat{P}\zeta(2)$, and PSLQ on $(x, 1, \zeta(2))$ returns $P, \hat{P}$ as exact rationals. Certification: the relation residual must vanish to half the working precision; the recovery must be stable under a +80-digit rerun; and the pipeline must reproduce the exact printed values of [1] in the totally symmetric case,

$$Q_1 = 21, \quad P_1 = \frac{87}{4}, \quad \hat{P}_1 = \frac{101}{4}, \quad Q_2 = 2989, \quad P_2 = \frac{1190161}{384}, \quad \hat{P}_2 = \frac{344923}{96},$$

which it does. An incidental end-to-end validation appears in §5: two independent recoveries at distinct parameter points agree with the exact factorial-ratio identity $I'(\mathbf{g}\mathbf{a}) = I'(\mathbf{a}) \cdot \prod h_i(\mathbf{a})! / \prod h_i(\mathbf{g}\mathbf{a})!$ through a discrepancy of 7^2 .

3 The published inclusions are false as stated

Brown–Zudilin state (experimentally) the inclusion $d_{m_1 n} \cdots d_{m_5 n} I'(\mathbf{a} \cdot n) \in \mathbb{Z}\zeta(5) + \mathbb{Z}$, where $m_1 \geq \cdots \geq m_5$ are the five largest entries of $\mathbf{h}(\mathbf{a})$, together with a sharpening by the factor $\Phi_n = \prod_{p > \sqrt{m_1 n}} p^{\nu_p}$, and, in the totally symmetric case, the display $Q_n, d_n^2 d_{2n} \hat{P}_n, d_n^5 P_n \in \mathbb{Z}$.

Already at $n = 2$ in the totally symmetric case, with the paper's own printed values: $d_2^5 P_2 = 2^5 \cdot \frac{1190161}{384} = \frac{1190161}{12} \notin \mathbb{Z}$ (off by $2^2 \cdot 3$), and $d_2^2 d_4 \hat{P}_2 = 48 \cdot \frac{344923}{96} = \frac{344923}{2} \notin \mathbb{Z}$ (off by 2). The numerators are coprime to 6, so the failures are exact. Our recovered values agree with the printed ones, independently. Analogous failures occur at 3 of 19 small interior points in our first systematic sample, always 2-adically. The corrected statements are the subject of the next section.

4 The sharp correction: data and conjecture

Extend Brown–Zudilin's sharpening to every prime (legitimate, since the factorial-ratio identity above is exact p -adically):

$$D_\nu(\mathbf{a}, n) = \frac{\prod_{i=1}^5 d_{m_i n}}{\Phi_n^*}, \quad \Phi_n^* = \prod_{\text{all } p} p^{\nu_p}, \quad \nu_p = \max_{g \in \mathfrak{G}} \text{ord}_p \frac{\prod_{i \in \mathcal{F}} (h_i(\mathbf{a})n)!}{\prod_{i \in \mathcal{F}} (h_i(\mathbf{g}\mathbf{a})n)!}.$$

Conjecture 4.1. *For all admissible $\mathbf{a}$ and $n \geq 1$, and every prime p ,*

$$\text{ord}_p \text{den}(P_n) \leq \text{ord}_p D_\nu(\mathbf{a}, n) + \begin{cases} 2, & p = 2, \\ 1, & p = 3, \\ 0, & p \geq 5 \text{ (see §5)}, \end{cases}$$

i.e. $12 D_\nu(\mathbf{a}, n) I'(\mathbf{a} \cdot n) \in \mathbb{Z}\zeta(5) + \mathbb{Z}$, and the ceilings at 2 and 3 are attained.

Evidence. Over 200 certified cells (systematic random interior points with entries up to 7, growth series at selected points with $n \leq 8$, and the authors' record point): the 2-adic excess distribution over the first 164 cells is $\{-6:2, -5:1, -4:4, -3:9, -2:9, -1:20, 0:33, +1:40, +2:46\}$ — ceiling +2, attained 46 times, never exceeded; the 3-adic distribution is $\{-4:1, -3:3, -2:8, -1:43, 0:133, +1:30\}$ — ceiling +1, attained, never exceeded. In the totally symmetric case the 2-adic ceiling is attained at every computed n ($n = 1, \dots, 5$ and 8): $\text{den}(P_n) = 4 \cdot 2^{\text{ord}_2 d_n^5} \cdot (\text{odd part})$, and $\text{den}(\hat{P}_n) = 2 \cdot 2^{\text{ord}_2 d_n^2 d_{2n}} \cdot (\text{odd part})$ for $n \leq 5$. The odd parts show only slack (the 3-adic prediction over-pays at most points, which is why the 3-part of the correction is visible only where the 3-adic prediction is tight).

Growth. Along every measured series the excess/slack trajectory is bounded: e.g. the largest observed slack point $\mathbf{a} = (4, 3, 4, 4, 3, 2, 4, 2)$ has 2-adic slack 6, 5, 4, 5, 3 bits at $n = 1, \dots, 5$; points with attained ceiling at $n = 1, 2$ can relax into slack at $n = 3$. The conjectured bound—not any pointwise formula—is the stable object.

5 A boundary-prime anomaly, and orbit coherence

Exactly one sampled orbit violates the $p \geq 5$ clause: at $\mathbf{a} = (6, 2, 6, 1, 6, 4, 5, 5)$, $n = 1$ (where $m_1 = 7$ is prime and $\nu_7 = 1$), $\text{den}(P_1) = 2^7 \cdot 3^4 \cdot 5^2 \cdot 7$ exceeds the ν -sharpened prediction by one factor of 7. Since the factorial-ratio identity is exact, this *predicts* that the maximizing orbit partner $\mathbf{a}' = (6, 2, 5, 2, 4, 6, 6, 5)$ must violate even the raw (ν -free) law; direct audit confirms $\text{den}(P_1(\mathbf{a}')) = 2^6 \cdot 3^4 \cdot 5^2 \cdot 7^2$ against a d -product paying a single 7. Excesses are thus *orbit-coherent*: the correct invariant is an orbit-level anomaly divisor.

The effect is rare and non-persistent: a targeted scan of 40 fresh points satisfying the same trigger condition ($m_1 \in \{5, 7, 11, 13\}$ prime, $\nu_{m_1} \geq 1$) produced *no* further instance, and the anomalous orbit itself relaxes to slack (-2 at $p = 7$) at $n = 2$, at both partners coherently. We therefore state Conjecture 4.1's $p \geq 5$ clause as holding up to such isolated small- n , boundary-prime ($p = m_1$) fluctuations of size 1; characterizing them exactly is open.

6 The mechanism

Lemma 6.1 (elimination cost). *Let e_0, e_2 span a 2-dimensional $\mathbb{Q}$ -vector space and let L be a lattice in the dual with basis γ_1, γ_2 pairing as $\langle \gamma_1, (e_0, e_2) \rangle = (1, \zeta(2))$, $\langle \gamma_2, (e_0, e_2) \rangle = (0, (2\pi i)^2)$. Every nonzero $\delta \in L$ with $\langle \delta, e_2 \rangle = 0$ is an integer multiple of $\delta_0 = 24\gamma_1 + \gamma_2$; in particular $\langle \delta, e_0 \rangle \in 24\mathbb{Z}$.*

Proof. $a\zeta(2) + b(2\pi i)^2 = 0$ and $\zeta(2)/(2\pi i)^2 = -\frac{1}{24}$ force $b = a/24$, so $24 \mid a$. $\square$

If the true integral lattice is finer, with $\gamma_2/2^k$ integral, the cost drops to $24/2^k$. The linear form I' arises exactly by such an elimination: $\text{Ext}_{MT(\mathbb{Z})_{\mathbb{Q}}}^1(\mathbb{Q}(0), \mathbb{Q}(2)) = 0$ [5] splits the weight-4 part rationally, and the identity $\zeta^{\mathfrak{m}}(2) = -(\mathbb{L}^{\mathfrak{m}})^2/24$ is available motivically [6, §2.2]. Under two explicitly stated hypotheses—(H1) the integrand's de Rham coefficients are D_{ν} -controlled (the integral form of Brown–Zudilin's own sketched Barnes analysis), and (H2) a normalization of the integral Betti lattice—one obtains $24 D_{\nu} I' \in \mathbb{Z}\zeta(5) + \mathbb{Z}$, with the constant improving by the lattice index. The measured sharp constants then *measure* that index: $12 = 24/2$ for P (equivalently, cost 2 on the natural coefficient $2P$), and 2 for $\hat{P}$. To our knowledge these are the first experimentally measured integral-lattice invariants of periods of $\overline{\mathcal{M}}_{0,8}$; the literature contains the $\mathbb{Q}$ -theory [8, 5, 7] and an empirical $\{2, 3\}$ -clearing phenomenon for motivic period

expansions [9], but—so far as we can determine—no integral-lattice synthesis. The support $\{2, 3\}$ matches the torsion $K_3(\mathbb{Z}) \cong \mathbb{Z}/48$ [10] of the relevant splitting obstruction; we flag this as a consistency, not a derivation.

7 The totally symmetric case: concrete derivation and a support theorem

In the totally symmetric case $\mathbf{a} = (1, \dots, 1)$ the Barnes representation reduces to the reciprocal-sine kernel with rational part $r(s)r(t)C(s, t)$, $r(s) = \prod_{i=1}^n (s+i) / \prod_{j=n+1}^{2n+1} (s+j)^2$, whose partial-fraction data is fully explicit: $B_j = (-1)^n \binom{n+a}{a} \binom{n}{a}^2 / n!$ and $A_j = B_j (3H_a - H_{n+a} - 2H_{n-a})$ with $a = j - n - 1$ (verified exactly for $n \leq 6$). Two consequences, proved in the accompanying notes:

Theorem 7.1 (support). *In the totally symmetric case every prime dividing $\text{den}(P_n)$ is at most $2n + 1$, and no prime $p \geq 5$ exceeds the d_n^5 budget.*

Moreover the $\{2, 3\}$ -excess enters *only* through the Laurent coefficients $1, \frac{1}{6}, \frac{1}{90}, \dots$ of the sine kernel: the constant term P_n picks up exactly one doubled factor of the $\zeta(2)$ -coefficient $\frac{1}{6}$, i.e. the factor $12 = 2 \cdot 6$ —the concrete avatar of Lemma 6.1—while $\frac{1}{90}$ (which would inject a 5) provably does not contribute, matching Theorem 7.1 and the data. The remaining gap to a full proof of $12 d_n^5 P_n \in \mathbb{Z}$ is a classical-type d_n^5 -collapse bookkeeping for the structured sums above; it is verified exactly for $n \leq 5$.

8 Cross-family controls

The same measurement standard applied to the classical corpus shows its correction factors are uniformly *removable*, in contrast to §4: (i) Apéry $\zeta(3)$: $d_n^3 a_n \in \mathbb{Z}$ for $n \leq 50$ (the quoted $2d_n^3$ is slack; consistent with Rhin–Viola’s exact theorem [4], where the genuine 2 rides on the $\zeta(3)$ -coefficient parity, not the denominator). (ii) $\zeta(4)$: $d_n^4 v_n \in \mathbb{Z}$ for $n \leq 5$ in Zudilin’s derived series (the factor 2 retained in [3, Théorème 3] is slack there). (iii) Zudilin’s asymmetric $\zeta(5), \dots, \zeta(11)$ forms: computing the five coefficients $z_{j,n}$ exactly for $n \leq 5$ (apparently for the first time), the leading factor 2 retained even in the *conjectured* law [3, eq. (17.1)] is unnecessary, with ~ 30 bits of 2-adic margin; the well-poised parity cancellation and the proven inclusion of [3] serve as engine validations. These are finite verifications, not proofs; but they isolate the Brown–Zudilin family—the only construction here that performs a $\zeta(2)$ -elimination—as the only one whose correction is attained.

9 The worthiness supremum, and what the arithmetic cannot do

For completeness we record the analytic side. Implementing the worthiness exponent $\gamma(\mathbf{a})$ of [1] (validated to 8 decimals against all three published values, including every printed intermediate), a global search of the 7-dimensional projective cone —steepest ascent from the published point on integer and half-integer lattices, 120 random multi-starts, 60 basin hops, and an exhaustive

scan of all 31,710 projectively distinct arithmetic-progression points to height 160—finds the supremum

$$\sup_{\mathbf{a}} \gamma = 0.86597135\dots,$$

attained exactly at the published point $\mathbf{a} = (8, 16, 10, 15, 12, 16, 18, 13)$, whose symmetric coordinates $(20.5; 3.5, 4.5, \dots, 9.5)$ form an arithmetic progression. Combined with the bounded-slack measurements of §4, the construction’s arithmetic is now understood to bounded factors, and none of it moves the exponential budget: within this family the barrier to $\gamma > 1$ is structural.

10 Limitations and outlook

All sharpness and slack statements are finite verifications under the certification protocol of §2; Conjecture 4.1 is falsifiable by a single future cell. The mechanism of §6 is proved only in the elementary Lemma 6.1 plus the conditional reduction; hypothesis (H2) is one finite lattice computation (torsion-freeness of $H^*(\overline{\mathcal{M}}_{0,n}, \mathbb{Z})$ [11] should make it tractable), and (H1) is the integral form of the Barnes bookkeeping sketched in [1]. The measurement pipeline itself is portable, and we suggest its use as an *arithmetic screen* for any future candidate constructions: the true denominators of a proposed family can now be measured in hours rather than conjectured. In a companion development, Lemma 4 of [2] (the asymptotic core of the elementary proof that one of $\zeta(5), \dots, \zeta(25)$ is irrational) has been fully formalized in Lean 4/Mathlib, with a complete formalization of the theorem in progress; details will appear separately.

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